# A LIMIT-PRESERVING BASE WITH LOCALLY PRESENTABLE FIBRES IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

*Editorial note.* The footnote '(details omitted)' covers the set-theoretic size check of the glueing lemma; retained verbatim and flagged for the reviewer.

*Editorial note.* Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 824 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

*Allowed standard prerequisites (editorial).*

- Affines are limits of finite presentation. algebra-lemma-ring-colimit-fp (cross-chapter, named). Writes V = lim V_i in Case I.
- Stackiness of fibre products. stacks-lemma-2-product-stacks-in-groupoids (same chapter, named). Gives Zariski stackiness of the fibre product in Case II.
- Glueing sheaves and spaces. bootstrap-lemma-glueing-sheaves, spaces-lemma-glueing-algebraic-spaces (cross-chapter, named). Reassembles Z from its local pieces.

*Source reference.* Original reference 00QN: focused-seeds/geometry/S9-crit-check-representable-limit-preserving/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 02ZX: focused-seeds/geometry/S9-crit-check-representable-limit-preserving/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 07WE: focused-seeds/geometry/S9-crit-check-representable-limit-preserving/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04T6: focused-seeds/geometry/S9-crit-check-representable-limit-preserving/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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*Date*: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original reference 000R: focused-seeds/geometry/S9-crit-check-representable-limit-preserving/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 02WQ: focused-seeds/geometry/S9-crit-check-representable-limit-preserving/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 02WR: focused-seeds/geometry/S9-crit-check-representable-limit-preserving/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 02ZL: focused-seeds/geometry/S9-crit-check-representable-limit-preserving/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Let $\kappa = size(T)$ for some $T \in \mathrm{Ob}((Sch/S)_{fppf})$. Let $f : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$ such that*

1. *$\mathcal{Y} \to (Sch/S)_{fppf}$ is limit preserving on objects,*
2. *for an affine scheme $V$ locally of finite presentation over $S$ and $y \in \mathrm{Ob}(\mathcal{Y}_V)$ the fibre product $(Sch/V)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X}$ is representable by an algebraic space of size $\leq \kappa$[1],*
3. *$\mathcal{X}$ and $\mathcal{Y}$ are stacks for the Zariski topology.*

*Then $f$ is representable by algebraic spaces.*

*Proof.* Let $V$ be a scheme over $S$ and $y \in \mathcal{Y}_V$. We have to prove $(Sch/V)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X}$ is representable by an algebraic space.

Case I: $V$ is affine and maps into an affine open $\mathrm{Spec}(\Lambda) \subset S$. Then we can write $V = \lim V_i$ with each $V_i$ affine and of finite presentation over $\mathrm{Spec}(\Lambda)$, see Algebra, Lemma 00QN. Then $y$ comes from an object $y_i$ over $V_i$ for some $i$ by assumption (1). By assumption (3) the fibre product $(Sch/V_i)_{fppf} \times_{y_i,\mathcal{Y}} \mathcal{X}$ is representable by an algebraic space $Z_i$. Then $(Sch/V)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X}$ is representable by $Z \times_{V_i} V$.

Case II: $V$ is general. Choose an affine open covering $V = \bigcup_{i \in I} V_i$ such that each $V_i$ maps into an affine open of $S$. We first claim that $\mathcal{Z} = (Sch/V)_{fppf} \times_{y,\mathcal{Y}} \mathcal{X}$ is a stack in setoids for the Zariski topology. Namely, it is a stack in groupoids for the Zariski topology by Stacks, Lemma 02ZL. Then suppose that $z$ is an object of $\mathcal{Z}$ over a scheme $T$. Denote $g : T \to V$ the morphism corresponding to the projection of $z$ in $(Sch/V)_{fppf}$. Consider the Zariski sheaf $I = Isom_{\mathcal{Z}}(z, z)$. By Case I we see that $I|_{g^{-1}(V_i)} = *$ (the singleton sheaf). Hence $\mathcal{I} = *$. Thus $\mathcal{Z}$ is fibred in setoids. To finish the proof we have to show that the Zariski sheaf

[1]The condition on size can be dropped by those ignoring set theoretic issues.

$Z : T \mapsto \mathrm{Ob}(\mathcal{Z}_T)/\cong$ is an algebraic space, see Algebraic Stacks, Lemma 02ZX. There is a map $p : Z \to V$ (transformation of functors) and by Case I we know that $Z_i = p^{-1}(V_i)$ is an algebraic space. The morphisms $Z_i \to Z$ are representable by open immersions and $\coprod Z_i \to Z$ is surjective (in the Zariski topology). Hence $Z$ is a sheaf for the fppf topology by Bootstrap, Lemma 07WE. Thus Spaces, Lemma 02WR applies and we conclude that $Z$ is an algebraic space[2]. □

[2]To see that the set theoretic condition of that lemma is satisfied we argue as follows: First choose the open covering such that $|I| \leq \mathrm{size}(V)$. Next, choose schemes $U_i$ of size $\leq \max(\kappa, \mathrm{size}(V))$ and surjective étale morphisms $U_i \to Z_i$; we can do this by assumption (2) and Sets, Lemma 04T6 (details omitted). Then Sets, Lemma 000R implies that $\coprod U_i$ is an object of $(Sch/S)_{fppf}$. Hence $\coprod Z_i$ is an algebraic space by Spaces, Lemma 02WQ.